

Klassik usullar bilan matnlarni tasniflash

Tabiiy tilni qayta ishlash — 2-ma'ruza

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11:00–12:20 · mavzu №2

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Takrorlash: L1 natijalarini tekshiramiz

Savol 1

D_1 ="nlp qiziq", D_2 ="python foydali",
 D_3 ="nlp foydali"

TF-IDF(qiziq, D_1) =?
[df = 1, N = 3, TF = 1]

Savol 2

BoW vektori [1, 1, 0, 0] nimani **saqlaydi**?
Nima **yo'qoladi**?

Takrorlash: L1 natijalarini tekshiramiz

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Savol 2

BoW vektori [1, 1, 0, 0] nimani **saqlaydi**?
Nima **yo'qoladi**?

Javob 1

IDF = $\ln(3/1) \approx 1.099$
TF-IDF = $1 \times 1.099 = \mathbf{1.099}$

Javob 2

Saqlaydi: so'zlar **chastotasi**.
Yo'qoladi: so'z **tartibi** (word order).
Bugun: shu vektorlar asosida tasniflash qilamiz.

Bugungi maqsadlar

- ➊ **Keltirib chiqara olasiz:** logistik regressiya gradienti — $\sigma(z) \rightarrow \ell_i \rightarrow L \rightarrow \partial L / \partial w_j$ — uch asosiy bosqichda.
- ➋ **Hisoblay olasiz:** Naive Bayes posterior nisbatini Laplace silliqlash (Laplace smoothing) bilan.
- ➌ **Qurasiz:** confusion matrix dan accuracy, precision, recall va F_1 metrikalarini.
- ➍ **Tanqidiy baho bera olasiz:** sentiment klassifikatorida kamida uch etika muammosini aniqlaysiz.

Minimum talablar: TF-IDF vektori (L1); Python funksiyalar; $\exp(\cdot)$ va $\ln(\cdot)$ tushunchalari.

Dars rejasi: to'rt blok

Blok	Mavzu	Vaqt
1	Logistik regressiya: sigmoid $\rightarrow$ gradient	22 daq
2	Naive Bayes: Bayes teoremasi + Laplace silliqdash	25 daq
3	Metrikalar: confusion matrix, precision, recall, F_1	20 daq
4	NLPda etika: uch muammo, oqibatlar	10 daq
Xulosa va P2 ko'prigi		3 daq

Amaliyotdagi misollar o'zbek matnlari asosida, CPU muhitida ham bajariladigan tarzda beriladi.

Bugungi darsda yechiladigan muammo

L1 da to'xtagan vaziyatimiz:

Sharh: "Sifati **yaxshi**, ammo yetkazib berish xizmati sifati **past** edi."

TF-IDF ning har bir so'z uchun qiymati:

yaxshi: 0.51 past: 0.38 sifat: 0.44 ammo: 0.22

Sodda qoidaga asoslangan model: "yaxshi" bor $\Rightarrow$ ijobiy sharh?

Xato! "ammo past" kontentini yetarlicha baholamaydi.

TF-IDF vektor $\mathbf{x}$



Matematik model



$P(\text{ijobiy} \mid \mathbf{x})$



Qaror (> 0.5 ?)

Bugun ikkita modelni qarab chiqamiz:

1. Logistik regressiya
2. Naive Bayes

Logistik regressiya: nima uchun ehtimollik?

Nima uchun ehtimollikka muhtojmiz?

“Ikki qiymat” yetarli emas:

“Zo'r mahsulot!”	ijobiy
“Mahsulot yaxshi ammo...”	noaniq?
“Yomon edi.”	salbiy

Model ikkinchi qatordagi **noaniqlikni** ifodalashi kerak.

$P(\text{ijobiy}) = 0.95$ — ishonchli

$P(\text{ijobiy}) = 0.53$ — noaniq

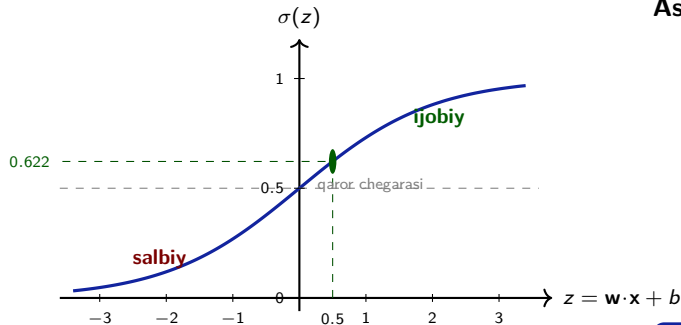
$P(\text{ijobiy}) = 0.08$ — salbiy

Qaror chegarasini turli vazifalarga moslashtirish:

spam filtri: > 0.90

tibbiy skrining: validatsiya asosida pastroq chegara

Sigmoid funksiyasi: z ni ehtimollikka aylantiradi



Asosiy xossalar:

- $\sigma(z) \in (0, 1)$ — har doim ehtimollik
- $\sigma(0) = 0.5$ — qaror chegarasi
- $z \gg 0$: $\sigma \rightarrow 1$ (ishonchli ijobiy)
- $z \ll 0$: $\sigma \rightarrow 0$ (ishonchli salbiy)

$$z = 0.8(1.0) + (-1.2)(0.0) + (-0.3) = 0.5$$

$$\sigma(0.5) \approx \mathbf{0.622}$$

$$0.622 > 0.5 \Rightarrow \mathbf{ijobiy}$$

$$P(y=1 \mid \mathbf{x}) = \sigma(\mathbf{w} \cdot \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w} \cdot \mathbf{x} + b)}}$$

bunda: $\mathbf{x}$ = kirish vektori (TF-IDF natijasi); $\mathbf{w}$ = vaznlar (weights); b = siljish had (bias); σ = sigmoid; $y=1$ = ijobiy sinf

Xossalar:

- $\sigma(z) \in (0, 1)$ — har doim ehtimollik (0,1) oraliqda
- $\sigma(0) = 0.5$ — qaror chegarasi
- $w_j > 0$: j -so'zni ijobiy tomonga siljitadi
- $w_j < 0$: salbiy tomonga

Misol (ikkita so'z):

$w_{\text{yaxshi}} = +0.8$ — ijobiy signal

$w_{\text{yomon}} = -1.2$ — salbiy signal

$b = -0.3$ — boshlang'ich noaniqlik

Qaror: $\sigma(\mathbf{w} \cdot \mathbf{x} + b) > 0.5?$

LR ni keltirib chiqarish, 1-qism: loss funksiyasi

1-qadam. Bitta namuna ehtimolligi ($z_i = \mathbf{w} \cdot \mathbf{x}_i + b$):

$$P(y_i | \mathbf{x}_i) = \sigma(z_i)^{y_i} (1 - \sigma(z_i))^{1-y_i}$$

$$(y_i=1: \sigma(z_i)^1; \quad y_i=0: (1 - \sigma(z_i))^1)$$

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($y_i=1$: $\sigma(z_i)^1$; $y_i=0$: $(1 - \sigma(z_i))^1$)

2-qadam. N ta namuna uchun log-likelihood:

$$\ell(\mathbf{w}, b) = \sum_{i=1}^N [y_i \ln \sigma(z_i) + (1 - y_i) \ln(1 - \sigma(z_i))]$$

(ko'paytmani yig'indiga aylantirib underflow oldini olamiz.)

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2-qadam. N ta namuna uchun log-likelihood:

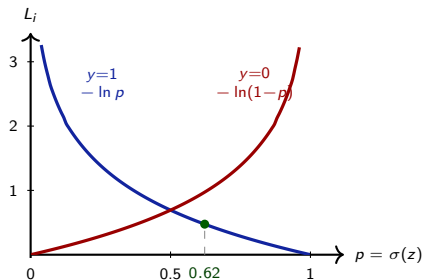
$$\ell(\mathbf{w}, b) = \sum_{i=1}^N [y_i \ln \sigma(z_i) + (1 - y_i) \ln(1 - \sigma(z_i))]$$

(ko'paytmani yig'indiga aylantirib underflow oldini olamiz.)

3-qadam. Cross-entropy loss:

$$L(\mathbf{w}, b) = -\frac{1}{N} \ell(\mathbf{w}, b)$$

($\max \ell \Leftrightarrow \min L$; manfiy ishora standart konvensiya)



Ishonchli **xato** $\Rightarrow L$ portlaydi; to'g'ri $\Rightarrow L \rightarrow 0$.

LR ni keltirib chiqarish, 2-qism: gradient va uni yangilash qoidasi

4-qadam. $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ dan foydalanib:

$$\frac{\partial L}{\partial w_j} = \frac{1}{N} \sum_{i=1}^N (\sigma(z_i) - y_i) x_{ij}$$

intuitiv o'qilishi: "bashorat xatosi" $\times$ "j-feature"

LR ni keltirib chiqarish, 2-qism: gradient va uni yangilash qoidasi

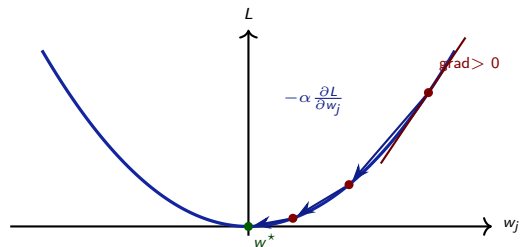
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intuitiv o'qilishi: "bashorat xatosi" $\times$ "j-feature"

5-qadam. SGD orqali yangilash (α = o'rganish tezligi):

$$w_j \leftarrow w_j - \alpha \frac{\partial L}{\partial w_j}$$



Har qadam w_j ni L minimumi sari suradi.

SGD mantig'i

$\sigma(z_i) > y_i$ (ortiqcha baho) $\Rightarrow \text{grad} > 0 \Rightarrow w_j \downarrow$.

$\sigma(z_i) < y_i$ (kam baho) $\Rightarrow \text{grad} < 0 \Rightarrow w_j \uparrow$.

LR ni qo'lda hisoblash

$\mathbf{x} = [\text{TF-IDF}(\text{yaxshi})=1.0, \text{TF-IDF}(\text{yomon})=0.0]; \quad \mathbf{w} = [0.8, -1.2], \quad b = -0.3$

$$z = 0.8(1.0) + (-1.2)(0.0) + (-0.3) = 0.5$$

$$\sigma(0.5) = \frac{1}{1 + e^{-0.5}} \approx \frac{1}{1.607} \approx 0.622$$

$$0.622 > 0.5 \Rightarrow \text{ijobiy} \checkmark$$

$$L_i = -\ln(0.622) \approx 0.475$$

Noto'g'ri bashorat: L_i oshadi.

To'g'ri va ishonchli: L_i kamayadi.

Eslatma: $e^{-0.5} \approx 0.607$; kodda `from scipy.special import expit; expit(0.5)`.

Sizning vazifangiz: LR bashoratini hisoblang

Vazifa (3 daqiqa)

Xuddi shu parametrlar: $\mathbf{w} = [0.8, -1.2]$, $b = -0.3$.

Yangi so'rov: $\mathbf{x} = [0, 1]$ (faqat "yomon" so'zi uchraydi).

a) $z = ?$ b) $\sigma(z) \approx ?$ ($e^{1.5} \approx 4.48$) c) Bashorat?

Sizning vazifangiz: LR bashoratini hisoblang

Vazifa (3 daqiqa)

Xuddi shu parametrlar: $\mathbf{w} = [0.8, -1.2]$, $b = -0.3$.

Yangi so'rov: $\mathbf{x} = [0, 1]$ (faqat "yomon" so'zi uchraydi).

a) $z = ?$ b) $\sigma(z) \approx ?$ ($e^{1.5} \approx 4.48$) c) Bashorat?

Javob

a) $z = 0.8(0) + (-1.2)(1) + (-0.3) = -1.5$

b) $\sigma(-1.5) = \frac{1}{1 + e^{1.5}} \approx \frac{1}{5.48} \approx 0.182$

c) $0.182 < 0.5 \Rightarrow$ **salbiy** ✓ ("yomon" so'zi salbiy og'irlikka ega: $w_2 = -1.2$)

Logistik regressiya kodi

```
from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

texts = ["yaxshi mahsulot",
         "ajoyib sifat",
         "yomon yetkazish"]
labels = [1, 1, 0]

pipe = Pipeline([
    ('tfidf', TfidfVectorizer()),
    ('lr', LogisticRegression(
        C=1.0, # teskari C
        max_iter=200)),
])
pipe.fit(texts, labels)
p = pipe.predict_proba(["yaxshi mahsulot"])[0][1]
print(f"P(ijobiy) = {p:.3f}")
```

Kod	Formula
TfidfVectorizer	$\mathbf{x}$
C=1.0	teskari C
max_iter	iteratsiyalar soni
predict_proba	$\sigma(\mathbf{w} \cdot \mathbf{x} + b)$
predict	$> 0.5?$

Muhim:

C kichik bo'lsa $\Rightarrow$
ko'proq regularizatsiya
(vaznlar kichrayadi)

LR da sinflar nomutanosibligi (class imbalance)

Muammo

O'quv ma'lumotlari: 900 salbiy, 100 ijobiy sharh.

Oddiy LR modeli deyarli barcha namunalarni “salbiy” deb bashorat qilishni o'rganishi mumkin.

Natija: accuracy=90%, lekin F1=0% — butunlay foydasiz model!

Yechim: `class_weight='balanced'`

```
LogisticRegression(class_weight='balanced')
```

Avtomatik og'irlik: $w_c = \frac{N}{K n_c}$

N — jami namunalar, K — sinflar soni, n_c — c sinfidagi namunalar soni.

Kam sinfga ko'proq og'irlik beriladi; macro F_1 validatsiyada tekshiriladi.

Alternativ usul: oversampling yoki threshold tuning.

Naive Bayes: sanashga asoslangan ehtimollik modeli

Savol: “yaxshi film” ijobiy sharh deganimi?

NB javobi: O'quv ma'lumotlarida “yaxshi” so'zi ijobiy matnlarda *necha marta* uchragan? Salbiy matnlarda-chi?

→ sanab chiqamiz → ehtimollikni hisoblaymiz → taqqoslaymiz

LR: $P(y|\mathbf{x})$ — qaror chegarasini to'g'ridan-to'g'ri o'rganadi
(diskriminativ model)

NB: $P(\mathbf{x}|y)P(y)$ — har sinfni modellaydi
(generativ model)

Afzalliklari:

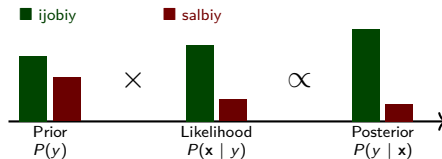
	LR	NB
O'qitish	iterativ	sanash orqali
Kichik data	regularizatsiya kerak	barqaror
Talqin	w_j	$P(w y)$

O'zbek tili uchun:

kichik korpusda NB barqaror natija beradi,
korpus hajmi o'sganda LR ham yaxshilanadi.

Naive Bayes: Bayes teoremasi va uning sodda farazi

$$P(y | \mathbf{x}) \propto P(\mathbf{x} | y) P(y)$$



Dalil (so'zlar) priorni qayta tortadi $\Rightarrow$ posterior keskinlashadi.

$$P(\mathbf{x} | y) \propto \prod_{j=1}^{|V|} P(w_j | y)^{x_j}$$

x_j — w_j ning sanog'i; so'zlar sinf berilganda shartli mustaqil deb faraz qilinadi (amalda ko'pincha buziladi).

bunda: $P(y)$ = prior; $P(\mathbf{x}|y)$ = likelihood; $P(y|\mathbf{x})$ = posterior; $|V|$ = lug'at hajmi

NB ni keltirib chiqarish, 1-qism:

1-qadam. NB tasniflash qoidasi:

$$\hat{y} = \arg \max_y \left[P(y) \prod_{j=1}^{|V|} P(w_j | y)^{x_j} \right]$$

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2-qadam. Logarifmga o'tish (monoton $\Rightarrow \arg \max$ o'zgarmaydi):

$$\hat{y} = \arg \max_y \left[\log P(y) + \sum_{j=1}^{|V|} x_j \log P(w_j | y) \right]$$

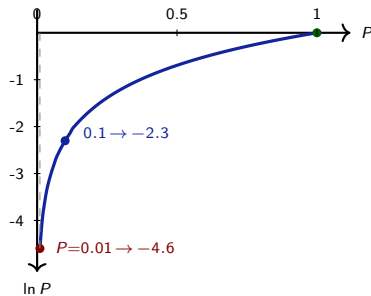
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Kichik $P \in (0, 1) \rightarrow$ chekli manfiy son; ko'paytma
 $\Rightarrow$ yig'indi.

Nima uchun log?

100 ta so'z, har biri $P \approx 0.01$:

$0.01^{100} = 10^{-200}$ — Python da 0.0 (*underflow* xavfi!)

Log bilan: $100 \times \log(0.01) = -460$ — xavfsiz.

NB ni keltirib chiqarish, 2-qism: Laplace (Add- α) silliqlash

0 marta uchrash muammosi

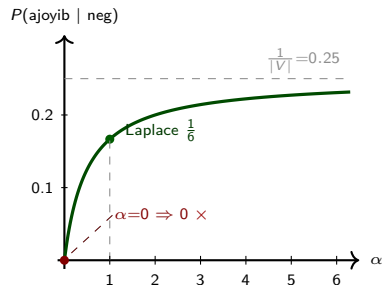
“ajoyib” so’zi *salbiy* sinfda 0 marta uchrasa:

$P(\text{ajoyib}|\text{neg}) = 0 \Rightarrow$ butun ko’paytma = 0 $\Rightarrow$ model ishlamaydi!

$$P(w | y) = \frac{\text{count}(w, y) + \alpha}{N_y + \alpha |V|}$$

$\alpha=0$ silliqlashsiz; $\alpha=1$ Laplace; $\alpha \gg 1$ barcha so’zlar teng.

bunda: $\text{count}(w, y)$ = w uchrashlar soni; N_y = y sinfdagi tokenlar; $|V|$ = lug’at hajmi; $\alpha=1$ = Laplace



α nolni ko'taradi; juda katta α teng taqsimotga itaradi.

Naive Bayes asosida tasniflashni qo'lda hisoblaymiz

O'quv to'plami: pos=[“yaxshi film”, “ajoyib film”], neg=[“yomon film”]

$|V|=4$ {yaxshi, ajoyib, yomon, film}; $P(\text{pos})=2/3$, $P(\text{neg})=1/3$

Laplace ($\alpha=1$):

$P(w y)$	pos ($N_y=4$)	neg ($N_y=2$)
film	$(2+1)/(4+4) = 3/8$	$(1+1)/(2+4) = 1/3$
yaxshi	$(1+1)/(4+4) = 1/4$	$(0+1)/(2+4) = 1/6$

1/6: nol chastota $\rightarrow$ Laplace natijasi

Test: “yaxshi film”

$$s_{\text{pos}} = \frac{2}{3} \cdot \frac{1}{4} \cdot \frac{3}{8} = \frac{6}{96} = \frac{1}{16} \approx 0.0625$$

$$s_{\text{neg}} = \frac{1}{3} \cdot \frac{1}{6} \cdot \frac{1}{3} = \frac{2}{108} = \frac{1}{54} \approx 0.0185$$

$$\text{nisbat} = \frac{1/16}{1/54} = \frac{54}{16} = \frac{27}{8} = \mathbf{3.375} \rightarrow \mathbf{\text{ijobiy}}$$

2-amaliyotda assert: $\text{abs}(\text{ratio}-3.375) < 0.01$ ga
dush kelasiz

Sizning vazifangiz: Naive Bayes asosida tasniflashni qo'lda hisoblang

Vazifa (4 daqiqa)

Xuddi shu korpus (hozirgina ko'rgan misolimiz). “**yomon**” so'zi (bitta so'z, $\alpha=1$) uchun Naive Bayes orqali quyidagilarni hisoblang:

- a) $P(\text{yomon}|\text{pos}) = ?$ b) $P(\text{yomon}|\text{neg}) = ?$
- c) $\text{score}_{\text{pos}}$ va $\text{score}_{\text{neg}}$. d) nisbat < 1 yoki > 1 ? Qaysi sinf?

Sizning vazifangiz: Naive Bayes asosida tasniflashni qo'lda hisoblang

Vazifa (4 daqiqa)

Xuddi shu korpus (hozirgina ko'rgan misolimiz). “**yomon**” so'zi (bitta so'z, $\alpha=1$) uchun Naive Bayes orqali quyidagilarni hisoblang:

- a) $P(\text{yomon}|\text{pos}) = ?$ b) $P(\text{yomon}|\text{neg}) = ?$
- c) $\text{score}_{\text{pos}}$ va $\text{score}_{\text{neg}}$. d) nisbat < 1 yoki > 1 ? Qaysi sinf?

Javob

- a) $P(\text{yomon}|\text{pos}) = (0+1)/(4+4) = 1/8$ (nol chastota!)
- b) $P(\text{yomon}|\text{neg}) = (1+1)/(2+4) = 2/6 = 1/3$
- c) $s_{\text{pos}} = (2/3)(1/8) = 1/12 \approx 0.083$; $s_{\text{neg}} = (1/3)(1/3) = 1/9 \approx 0.111$
- d) nisbat $= (1/12)/(1/9) = 9/12 = 3/4 = 0.75 < 1 \Rightarrow$ **salbiy** ✓

Naive Bayesni kodd ishlatish

```
from sklearn.feature_extraction.text import (
    CountVectorizer)
from sklearn.naive_bayes import MultinomialNB

korpus = [
    "yaxshi film", # ijobiy (1)
    "ajoyib film", # ijobiy (1)
    "yomon film",  # salbiy (0)
]
labels = [1, 1, 0]

vec = CountVectorizer()
X = vec.fit_transform(korpus)

nb = MultinomialNB(alpha=1.0) # Laplace
nb.fit(X, labels)

test = vec.transform(["yaxshi film"])
proba = nb.predict_proba(test)
# proba[0] = [P(salbiy), P(ijobiy)]
print(f"P(ijobiy) = {proba[0][1]:.4f}")
```

Kod	Formula
alpha=1.0	α (Laplace)
fit_transform	$\text{count}(w, y), N_y$
predict_proba	$\log \sum \rightarrow \exp$
predict	$\arg \max$

Tekshirib ko'ring:

nb.feature_log_prob_
 $\rightarrow \log P(w|y)$ matritsasi natijasini beradi

Nisbatni olish:

np.exp(
 log_sp - log_sn)

Naive Bayesda korrelyatsiyalangan xususiyatlar muammosi

Shartli mustaqillik farazi buzilish muammosi:

“juda yaxshi” → “juda” va “yaxshi” birga uchraydi.

Naive Bayes (NB) ikkisini **mustaqil** hisoblaydi → ijobiy sinf uchun hisoblash jarayonida yaxshi so'zi **ikki marta** sanaladi, model ehtimollikni sun'iy ravishda oshirib yuboradi va ortiqcha ishonchli bashoratga olib keladi.

“Sifati yaxshi, sifatli mahsulot” jumlasida bir ildizga yaqin so'zlar model signalini takrorlashi mumkin.

Yechimlar

① Binary xususiyatlar:

`CountVectorizer(binary=True)`

② Logistik regressiya bilan bashorat qilish: u korrelyatsiyalangan belgilarga odatda bardoshliroq.

train-test to'plamlarida ma'lumotlar sizib o'tishi (leakage)

Noto'g'ri tartib

- 1) Avval butun korpusda `TfidfVectorizer.fit()` qilish;
- 2) keyin train/testga bo'lish.

Natijada, test ma'lumotidagi so'zlar umumiy ma'lumot ichida IDFGa kirib ketadi, model test ma'lumotlari taqsimotini oldindan ko'radi va model aniqligi sun'iy oshadi.

To'g'ri tartib

```
train_test_split(..., stratify=y) → Pipeline([('tfidf', TfidfVectorizer()), ('clf', model)]) → pipe.fit(X_train, y_train) → pipe.predict(X_test).
```

Amaliy qoida: preprocessing va vektorizatsiya parametrlari **faqat train** to'plamdan o'rganiladi.

2-amaliyotda Logistik regressiya (LR) va Naive Bayes (NB) aynan Pipeline ichida solishtiriladi.

Model natijasini baholash uchun nima uchun accuracy yetarli emas?

Muammo: 1000 sharh, 900 salbiy, 100 ijobiy.

Model A sharhlarni o'rganish jarayonida aksariyat holda salbiy sharhni ko'rdi va u yangi sharhni testlashda *“salbiy” deb tasniflash ehtimoli yuqori.*

$\text{Accuracy} = 900/1000 = \mathbf{90\%}$

Lekin: 100 ijobiy sharhning **birortasini** to'g'ri topmadi!

Kompaniyaning “90% aniqlik”da ishlaydigan modeli aslida ijobiy sharh yozgan mijozlarga g'alati ko'rinadi.

Raqamlar aldaydi — bizga **muvozanatli** metrika kerak.

Confusion matrix va to'rt metrika

	Bashorat +	Bashorat -
Haqiqat - Haqiqat +	TP to'g'ri	FN xato
Haqiqat - Haqiqat -	FP xato	TN to'g'ri

■ diagonal = to'g'ri bashorat (TP, TN)

■ tashqarisi = xato (FP — noto'g'ri ijobiy; FN — o'tkazib yuborilgan)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F_1score = \frac{2PR}{P + R} = \frac{2TP}{2TP + FP + FN}$$

Precision va recall: qachon qaysi biri muhim?

Precision: “ijobiy deganlarimizning qanchasi haqiqatan ijobiy?”

Spam filtri modeli qursangiz va FP yuqori bo'lsa, bu yomon (muhim xat spam deb belgilanadi).

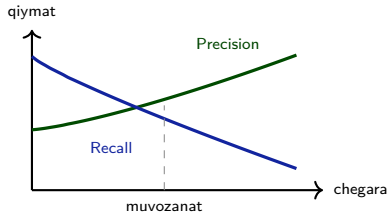
Recall: “Haqiqiy ijobiylarning qanchasini topdim?”

Tibbiy diagnoz qiluvchi model qursangiz va FN yuqori bo'lsa, bu yomon (kasallikni o'tkazib yuborish).

Savdoning ikki tomoni

Qaror chegarasi oshsa odatda $P \uparrow$, $R \downarrow$; chegara pasaysa aksincha.

F_1 — precision va recallning **garmonik o'rtachasi**; ikkalasini birga baholaydi.



F_1 score

Nima uchun garmonik, arifmetik emas?

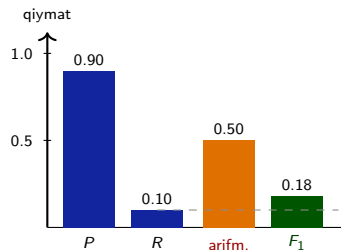
$P=0.90$, $R=0.10$ bo'lsa (model juda oz ijobiyini topgan):

Arifmetik: $(0.90 + 0.10)/2 = 0.50$

— o'rtacha (hammasi yaxshiday). **Aldaydi!**

Garmonik: $F_1 = \frac{2(0.90)(0.10)}{0.90 + 0.10} = \mathbf{0.18}$

— past natija ✓



Model	Accuracy	Precision	Recall	F_1
A (hammasi salbiy)	90%	—	0%	0%
B (o'rgangan; TP=60, FP=80, FN=40)	88%	43%	60%	≈50%

Model A accuracy yuqori, lekin $F_1=0$ — butunlay foydasiz!

Confusion matrixdan metrikalarni qo'lda hisoblab olish

Model B: TP=60, TN=820, FP=80, FN=40 (jami 1000)

$$A = (60 + 820)/1000 = 0.88$$

$$P = 60/(60 + 80) = 60/140 \approx 0.429$$

$$R = 60/(60 + 40) = 60/100 = 0.600$$

$$F_1 = \frac{2(0.429)(0.600)}{0.429 + 0.600}$$

$$\approx 0.515/1.029 \approx \mathbf{0.500}$$

Xulosa

Accuracy=88% (A modelning 90% natijasidan past)

$F_1 = 50\%$ (A modelning 0% natijasidan baland)

Model B ancha foydali!

100 ijobiydan 60 tasi topildi (Recall=60%).
Qolgan 40 FN ni 2-amaliyotda yaxshilashni o'rganasiz.

Sizning vazifangiz: Model C metrikalari

Vazifa (4 daqiqa)

Model C natijalari (1000 namuna (sample): 900 salbiy, 100 ijobiy):

TP=80, TN=780, FP=120, FN=20

a) Accuracy =? b) Precision =? c) Recall =? d) $F_1 \approx ?$

Sizning vazifangiz: Model C metrikalari

Vazifa (4 daqiqa)

Model C natijalari (1000 namuna (sample): 900 salbiy, 100 ijobiy):

TP=80, TN=780, FP=120, FN=20

a) Accuracy =? b) Precision =? c) Recall =? d) $F_1 \approx ?$

Javob

a) $(80 + 780)/1000 = 860/1000 = \mathbf{86\%}$

b) $80/(80 + 120) = 80/200 = \mathbf{40\%}$

c) $80/(80 + 20) = 80/100 = \mathbf{80\%}$

d) $F_1 = 2(0.40)(0.80)/(0.40 + 0.80) = 0.64/1.20 \approx \mathbf{53\%}$ (B=50% dan biroz yuqori)

O'rgangan metrikalarni kodda ishlatish

```
from sklearn.metrics import (
    classification_report,
    confusion_matrix,
    ConfusionMatrixDisplay)

# y_true, y_pred -- model bashoratlari
report = classification_report(
    y_true, y_pred,
    target_names=['salbiy', 'ijobiy'])
print(report)

cm = confusion_matrix(y_true, y_pred)
disp = ConfusionMatrixDisplay(
    cm, display_labels=['salbiy', 'ijobiy'])
disp.plot()
```

Atamalar	Formula
precision	$TP/(TP + FP)$
recall	$TP/(TP + FN)$
f1-score	$2PR/(P + R)$
support	n_y
macro avg	teng $\bar{F}_1$
weighted	n_y -og'irlik

Tavsiya:

"Imbalance" sinf uchun

macro avg

F_1 ishlatilg!

macro vs weighted F_1

weighted F_1 — aldaydi

900 salbiy, 100 ijobiy.

Salbiy $F_1=0.95$, ijobiy $F_1=0.30$.

Weighted:

$(0.95 \times 900 + 0.30 \times 100) / 1000$
 $= 0.885$ — ajoyib ko'rinadi!

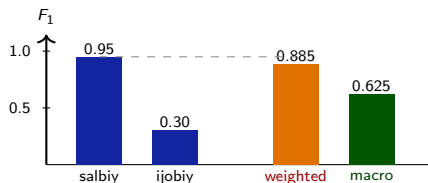
ijobiy sinf yomon bashoratlanayotgani
yashirin qoladi.

macro F_1 — to'g'riroq

Macro: $(0.95 + 0.30) / 2 = 0.625$

Kamroq uchraydigan sinf muammosi
ko'rinadi.

Qoida: sentiment tahlilida har ikki sinf
teng emasmi $\Rightarrow$ **macro** F_1 ishlatilg.



Etika — modelingiz kimga zarar yetkazishi mumkin?

Vaziyat:

Kompaniya sentiment modeli barcha viloyatlar mijozlari sharhlarini baholaydi.

Train ma'lumotlari asosan Toshkent sharhlaridan yig'ilgan.

Farg'ona shevasidagi so'zlar
→ lug'atda yo'q
→ “salbiy” standart bashorat
→ sistematik xato

Hududiy mijozlar fikri to'g'ri baholanmaydi

Kompaniya mahsulotini faqat bir mintaqa uchun optimallashtiradi

Adolat buziladi

1. Bir tomonga og'gan (biased) ma'lumot

O'quv ma'lumotlari bir tomonga og'sa, model shu og'ishni takrorlaydi. Masalan, 95% salbiy sharh bo'lsa, ijobiy o'zgarishlar "ko'rinmay" qoladi.

2. Kam resursli til va dialekt

Dialekt yoki kam uchraydigan shakllar "out-of-vocabulary" (OOV) bo'lsa, model sistematik xato qiladi. Natijada, ayrim mintaqa foydalanuvchilari xizmat yoki kredit tizimida zarar ko'radi.

3. Shaffoflik yetishmasligi

"Arizangiz rad etildi" degan natija sabab bilan izohlanmasa, apellyatsiya imkoni cheklanadi. Amaliy yechim: muhim so'zlar, ehtimollik, metrika va cheklovlarni hisobotda ko'rsatish.

Vaziyat (5 daqiqa)

Bank kreditiga ariza bergan shaxsning ijtimoiy tarmoq xabarlari sentiment modeli bilan baholanadi.

- 1 Model o'zbek dialektlarini tanimaydi.
- 2 O'quv ma'lumoti: 90% salbiy xabarlardan iborat.
- 3 Natija faqat "ijobiy/salbiy" — tushuntirish yo'q.

Savollar:

- a) Bu tizimda yuqoridagi uch etika muammosining qaysilari mavjud?
- b) Har bir muammo uchun bitta amaliy yechim taklif eting.
- c) Modelni "aniqroq" qilish (accuracy ni oshirish) etika muammosini hal qiladimi?

O'zbek tili va tasniflash xususiyatlari

Agglutativ til va BoW muammosi

“yaxshi” $\neq$ “yaxshiroq” $\neq$ “yaxshilik” — BoW da **uch alohida ustun**.

Test to'plamidagi yangi shakl so'zlar lug'atga kirmasa, vektorlashtirishda yo'qolib ketadi. So'z lug'atda bor, lekin sinfda uchramagan bo'lsa, NB da $P(w | y)=0$ xavfi paydo bo'ladi.

NB yondashuvi: Laplace lug'atdagi, ammo ayrim sinfda uchramagan so'zlarni nol ehtimollikdan saqlaydi.

Haqiqiy OOV uchun char n-gram yoki <UNK> kerak.

Yechim: preprocessing va n-gramlar

`preprocess()` “yaxshiroq”, “yaxshilik” kabi shakllarni yagona asosga yaqinlashtiradi.

Lug'at hajmi kamayadi, OOV xavfi pasayadi, lekin to'liq yo'qolmaydi.

LR va NB: qaysi birini tanlash kerak?

Mezon	Logistik regressiya	Multinomial NB
Tur	Diskriminativ	Generativ
O'qitish	Iterativ (SGD)	Bir marta, tez
Kichik korpus	regularizatsiya kerak	ko'pincha barqaror
Katta korpus	Kuchli	Yetarlicha
Korrelyatsiya	Bardoshli	Zaif (shartli mustaqillik muammosi)
Interpretatsiya	w_j koeffitsientlari	$P(w y)$
O'zbek morfologiyasi	stemming / n-gram kerak	stemming / n-gram kerak
Tengsiz sinf	class_weight, threshold	fit_prior, threshold

Amaliy qoida: kichik o'zbek korpus (<1000 namuna) → NB bilan baseline quring; keyin LR ni ham stratifikatsiyalangan validatsiyada sinab ko'ring.

O'qish uchun maqola: McCallum va Nigam (1998)

McCallum, A., Nigam, K. (1998).
A Comparison of Event Models for Naive Bayes Text Classification.
AAAI/ICML-98 Workshop on Learning for Text Categorization.

Asosiy topilma: Matn tasniflashda Multinomial NB ko'pincha Bernoulli NB dan ustun — so'zlar chastotasini hisobga olish muhim.

Shu sababli bu kursda `CountVectorizer + MultinomialNB(alpha=1.0)` qo'llanilgan.

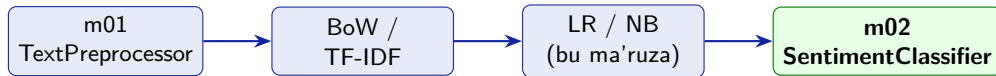
Muhokama savoli (2-amaliyotgacha o'ylang)

Kichik o'zbek korpus (500 namuna) uchun LR yoki NB qaysi yaxshi ishlaydi? Qanday eksperiment bilan aniqlaymiz?

Maslahat: kross-validatsiya + macro F_1

- 1 ✓ LR — $\sigma(z) \rightarrow \ell_i \rightarrow L \rightarrow \partial L / \partial w_j$ ni keltirib chiqardik.
- 2 ✓ NB posterior nisbati Laplace silliqdash bilan hisobladik.
- 3 ✓ Confusion matrix dan $A=88\%$, $P \approx 43\%$, $R=60\%$, $F_1 \approx 50\%$ qurdik.
- 4 ✓ Uch etika muammosi — og'gan ma'lumot, kam resurs tili, shaffoflik yo'qligiga tanqidiy baho berdik.

Ertangi amaliyotda sentiment klassifikatori qurasiz



2-amaliyotda:

- `SentimentClassifier(m02): fit() / predict() / predict_proba()`
- LR va NB ni macro F_1 bilan taqqoslash
- `confusion_matrix + classification_report` qurib ko'rasiz.

2-amaliyotning birinchi assert qiymati [28

```
-slayddan olinadi.] assert abs(ratio - 3.375) < 0.01  
# Ma'ruzaning [28]-slaydi bilan solishtiring
```

- ❶ McCallum, A., Nigam, K. (1998). *A Comparison of Event Models for Naive Bayes Text Classification*. AAAI/ICML-98 Workshop.
- ❷ Jurafsky, D., Martin, J.H. (2024). *Speech and Language Processing*, 3rd ed. §4 (Naive Bayes), §5 (Logistic Regression), §4.7 (Evaluation). web.stanford.edu/~jurafsky/slp3/
- ❸ Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *JMLR* 12, 2825–2830. `sklearn.naive_bayes`, `sklearn.linear_model`, `sklearn.metrics`
- ❹ Mitchell, M. et al. (2019). Model Cards for Model Reporting. *FAccT '19*. Etika va shaffoflik standarti.

Ilova S1: sigmoid xossalari va foydali farazlar

Ta'rif: $\sigma(z) = \frac{1}{1 + e^{-z}}$

Hosila:

$$\sigma'(z) = \frac{e^{-z}}{(1 + e^{-z})^2} = \sigma(z)(1 - \sigma(z))$$

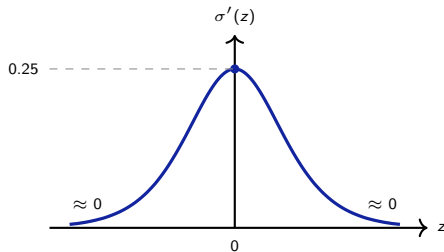
Isboti: $\frac{d}{dz}(1 + e^{-z})^{-1} = (-1)(1 + e^{-z})^{-2}(-e^{-z})$

Chegaralar:

$$\lim_{z \rightarrow +\infty} \sigma(z) = 1; \quad \lim_{z \rightarrow -\infty} \sigma(z) = 0; \quad \sigma(0) = 0.5$$

Qo'lda hisob uchun farazlar:

$$e^{-0.5} \approx 0.607; \quad e^{0.5} \approx 1.649; \quad e^{1.5} \approx 4.48; \quad e^{-1.5} \approx 0.223; \quad e^1 \approx 2.718$$



Gradient $z=0$ (qaror chegarasi) da *eng katta*; ishonchli bashoratda $\rightarrow 0$.

Ilova S2: Bayes teoremasini chiqarish

Birgalik ehtimollik:

$$P(y) \cdot P(\mathbf{x} | y) = P(y, \mathbf{x}) = P(\mathbf{x}) \cdot P(y | \mathbf{x})$$

Shundan:

$$P(y | \mathbf{x}) = \frac{P(\mathbf{x} | y) P(y)}{P(\mathbf{x})}$$

Klassifikatsiyada: $P(\mathbf{x})$ barcha y lar uchun bir xil $\Rightarrow$ proporsional yozish mumkin:

$$P(y | \mathbf{x}) \propto P(\mathbf{x} | y) P(y)$$

Shu sababli NB da maxraj ($P(\mathbf{x})$) hisoblash shart emas — faqat numeratorlarni taqqoslaymiz.

Ilova S3: F_β formulasi

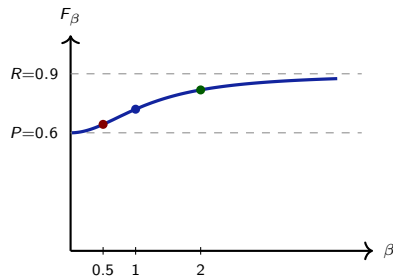
Agar Recall R Precision P dan muhimroq bo'lsa ($\beta > 1$):

$$F_\beta = \frac{(1 + \beta^2) P \cdot R}{\beta^2 P + R}$$

bunda: $\beta > 1$: recallga ko'proq; $\beta < 1$: precisionga ko'proq og'irlik

Misollar:

- Tibbiy diagnoz: $\beta = 2$ (FN qimmat)
 $F_2 = 5PR/(4P + R)$
- Spam filtri: $\beta = 0.5$ (FP qimmat)
 $F_{0.5} = 1.25PR/(0.25P + R)$
- Standart sentiment: $\beta = 1 \Rightarrow F_1 = 2PR/(P + R)$



β kichik $\rightarrow F \approx P$; β katta $\rightarrow F \approx R$.